Chapter 8 — Chemical Equilibrium

Complete Notes · PECTAA Chemistry 11 · Board-style longs + conceptual shorts

Topic 1 — Reversible reactions and dynamic equilibrium

Q.1 Explain reversible reactions, dynamic equilibrium and the difference between macroscopic and microscopic events.

A reversible reaction proceeds in both forward and reverse directions. In a closed system, the forward rate decreases as reactants are used up, while the reverse rate increases as products accumulate. Eventually the two rates become equal and the concentrations remain constant. This is dynamic equilibrium.

Macroscopically, properties such as colour, pressure and concentration appear constant. Microscopically, molecules continue to react in both directions. Equilibrium is therefore dynamic, not static.

Important Conceptual Short Questions — Topic 1

1. Why is equilibrium described as dynamic rather than static?

The forward and reverse reactions continue at equal rates. There is no net change in concentration, but molecular activity does not stop.

2. How do forward and reverse rates change as equilibrium is approached?

The forward rate decreases as reactant concentrations fall, while the reverse rate increases as product concentrations rise, until they become equal.

3. Why must equilibrium be studied in a closed system?

If reactants or products escape, the reverse reaction cannot balance the forward reaction and equilibrium cannot be established.

4. How can macroscopic constancy hide microscopic activity?

Observable properties remain constant because the rates of opposing processes are equal, even though individual particles are continuously reacting.

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Topic 2 — Equilibrium constant and its characteristics

Q.2 Derive the equilibrium constant expression and state the characteristics of K.

For aA + bB ⇌ cC + dD, the equilibrium constant is $K_c = [C]^c[D]^d/[A]^a[B]^b$. For gases, K_p uses partial pressures. Pure solids and pure liquids are omitted because their concentrations are effectively constant.

K depends only on temperature. It does not depend on initial concentrations, pressure or catalyst. A large K means products are favoured; a small K means reactants are favoured. A catalyst speeds up both forward and reverse reactions equally, so it helps equilibrium to be reached faster but does not change K or the equilibrium position.

Important Conceptual Short Questions — Topic 2

1. Why are pure solids and liquids omitted from equilibrium expressions?

Their concentrations are constant at a given temperature, so they are included in the constant value of K rather than written in the expression.

2. How does the magnitude of K indicate the position of equilibrium?

A large K means the numerator, product concentrations, is large compared with reactants. A small K means reactants predominate.

3. Why does a catalyst not change the equilibrium constant?

A catalyst lowers the activation energy of both forward and reverse reactions equally. It changes the rate of reaching equilibrium, not the relative stability of reactants and products.

4. Why does K change only with temperature?

Temperature changes the relative energies and rates of the forward and reverse processes, altering the equilibrium ratio. Concentration and pressure changes shift the position but not K.

Topic 3 — Le Chatelier's principle and effects of conditions

Q.3 State Le Chatelier's principle and explain the effects of concentration, pressure and temperature on equilibrium.

Le Chatelier's principle states that if a system at equilibrium is disturbed, it shifts in the direction that reduces the disturbance. Adding a reactant shifts equilibrium forward; adding a product shifts it backward. Removing a product shifts it forward.

Increasing pressure shifts a gaseous equilibrium toward the side with fewer moles of gas. Increasing temperature shifts equilibrium in the endothermic direction; decreasing temperature favours the exothermic direction. A catalyst does not shift equilibrium.

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Important Conceptual Short Questions — Topic 3

1. How does removing a product increase the yield of a reversible reaction?

Removing a product disturbs the equilibrium ratio. The system responds by forming more product, so the forward reaction is favoured.

2. Why does pressure affect only equilibria with a change in gas moles?

Pressure changes affect gases significantly. If both sides have the same number of gas moles, compression does not favour either side.

3. How does temperature affect an exothermic equilibrium?

Raising temperature favours the endothermic reverse reaction, reducing product yield. Lowering temperature favours the exothermic forward reaction.

4. Why does a catalyst not increase the equilibrium yield?

It increases the rates of both forward and reverse reactions equally, so the final equilibrium composition remains unchanged.

Topic 4 — Industrial applications of equilibrium

Q.4 Explain the industrial applications of equilibrium in the Haber and Contact processes.

In the Haber process, $N_2 + 3H_2 \rightleftharpoons 2NH_3$ is exothermic and proceeds with a decrease in gas moles. High pressure favours ammonia, while low temperature favours yield but slows the rate. A compromise temperature of about 400–450 °C, pressure of 200–300 atm and an iron catalyst are used.

In the Contact process, $2SO_2 + O_2 \rightleftharpoons 2SO_3$ is exothermic. High pressure and low temperature favour SO_3, but a compromise temperature of about 400–450 °C and a V_2O_5 catalyst are used to obtain a good rate and yield.

Important Conceptual Short Questions — Topic 4

1. Why is a compromise temperature used in the Haber process?

Low temperature gives a higher equilibrium yield but a very slow rate. A moderate temperature with a catalyst gives a reasonable rate and acceptable yield.

2. How does high pressure favour ammonia formation?

The forward reaction changes four moles of gas into two moles. High pressure shifts equilibrium toward the side with fewer gas moles, producing more NH_3.

3. Why is V_2O_5 used in the Contact process?

It catalyses the oxidation of SO_2 to SO_3, allowing a useful rate at a temperature where the equilibrium yield is still favourable.

4. How does recycling unreacted gases improve industrial yield?

Unreacted gases are passed again over the catalyst, so more reactant is converted to product over time, improving overall efficiency.

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